



Passenger Experience Analytics

A Data-Driven Approach to Airline Product Strategy

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01

Section

Data Analysis

Explore the data, understand the passenger, drive the product.



Portfolio Website & Power BI Dashboard



GitHub Repository



**Click Here to Access the
Dashboard!**

Analysis Overview

Six analytical workstreams. 1,957 real voices. Every output framed with a product decision in mind.

 1,957 Skytrax reviews

 May 2016 – May 2026

 6 analytical workstreams

- Scraped directly from **airlinequality.com** — no aggregator dependency
- Every output is framed around a **product decision**, not just a statistical result



Data Collection

1,957 Skytrax reviews scraped across a 10-year span from airlinequality.com

EDA



Python Scraper

Custom-built scraper to extract and structure passenger reviews at scale

Python



Six Workstreams

EDA · NLP · Semantic Clustering · Power BI · Synthetic A/B Tests · Propensity Model

NLP · ML



Product-Decision Framing

Every analytical output mapped to a commercial or product decision — actionable by design

Strategy

Data Overview

1,957 real voices. 10 years. 26 variables. Every data point genuine passenger voice.

1,957

Skytrax Reviews

26

Variables per Review

5.8/10

2024 Average Rating — Lowest on Record

119,500

Rows — 5 Synthetic Datasets



Primary Dataset — Skytrax
Scraped directly from airlinequality.com — not sourced from an aggregator. May 2016 – May 2026.



26 Variables per Review
Overall rating, 7 category scores, cabin class, route, aircraft, country, full review text.



5 Synthetic Operational Datasets
119,500 rows total — statistically realistic distributions with embedded anomalies for modelling.



2024 — Worst-Rated Year on Record
5.8/10 average — a 2.4-point fall from the 2021 peak. Trend demands urgent product intervention.



Value for Money — Strongest Predictor
Pearson $r = 0.86$ correlation with positive recommendation — the primary product investment lever.



First Class Food Gap
First Class food scores avg 3.87 vs Business Class 4.06 — a 0.19pt premium cabin expectation gap.



Wi-Fi Non-Response Rate: 46%
Passengers cannot rate what they cannot use — a direct product gap signal in connectivity.



Verified vs Unverified Reviews
Verified reviews score 1.1pts lower — a methodological anchor for interpreting raw Skytrax ratings.

Sentiment Analysis

What passengers say, and what they actually mean — grouping complaints by meaning, not by words.



LLM-Powered Sentiment Extraction

Llama 3.1 8B & Qwen via Groq API and Ollama extract context-aware sentiment — replaces keyword-matching. 40+ structured fields per review: score, emotion, sarcasm flag, competitor mentions, rebook intent.

✓ 40+ fields per review



Semantic Clustering by Meaning

Sentence embeddings (all-MiniLM-L6-v2, 384 dimensions) cluster complaints by semantic direction. Silhouette analysis tests k=4 to k=11 — no arbitrary grouping.

📄 384-dim embeddings



Product-Ready Cluster Breakdowns

Each cluster carries route, cabin, and time-period breakdowns — a product manager can act without reading individual reviews. 10 product theme dimensions output as Power BI-ready CSVs.

📄 10 product dimensions

PSI 65 vs Rec. Rate 71%

384-dim embeddings

Advocacy Gap Analysis

65

Passenger
Sentiment Index

71%

Recommendation
Rate



Advocacy Gap: 6 points — passengers recommend before they truly feel it



Raw complaint tables understate scale: "rude staff", "dismissive crew", "unhelpful attendant" are **one product failure** counted as three — semantic clustering corrects this.



5 Interactive Pages

The Big Picture · Passenger Experience · Product Analysis · Route & Geographic Intelligence



Custom DAX Measures

PSI · Advocacy Gap · YoY Rating Change · Recommendation
Propensity — built on the NLP pipeline output



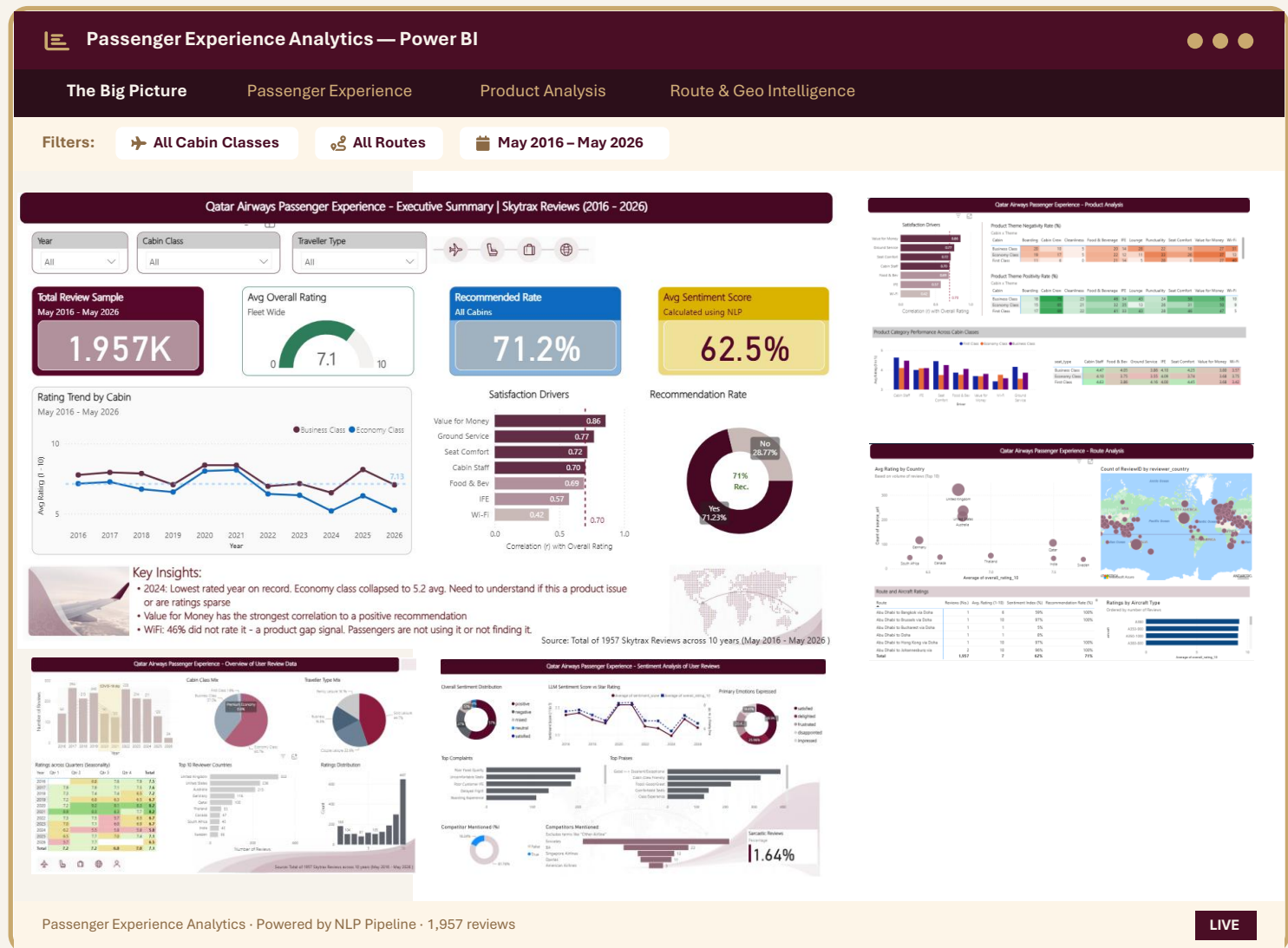
Route Filtering

Filter by cabin class, route, aircraft type & time period — commercial risk identification at a glance

5
DB Pages

4 DAX Measures

NLP Pipeline Fed



A/B Test: Elevated Meal Service

✓ High Confidence

The intervention works. Roll out with high confidence.

👤 n=6,000 passengers 🛫 4 long-haul routes 📅 4 months 📈 p<0.001 · Cohen's d = 0.609

Control Group

Standard Economy Meal

Satisfaction Score

3.607

Out of 5.0

Complaint Rate

8.0%

Meal-related complaints

Finish Rate

Baseline

Reference group

Economy Lift

—

No intervention applied

+0.30pt

Satisfaction Lift

−46%

Complaint Rate Drop

+10pp

Finish Rate Gain

Economy

Largest Lift Cabin

Treatment Group

Elevated Economy Meal

Satisfaction Score

3.906

Out of 5.0

Complaint Rate

4.3%

Meal-related complaints

Finish Rate

+10pp

vs. control group

Economy Lift

Largest

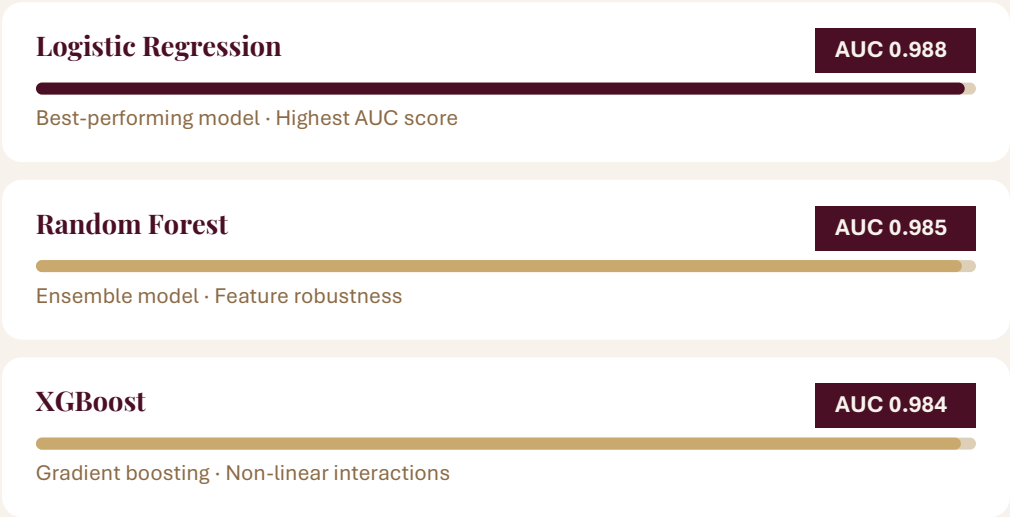
First Class at satisfaction ceiling

Tests Applied: t-test Mann-Whitney U Chi-squared Shapiro-Wilk Cohen's d = 0.609 95% CI on lift

Recommendation Propensity Model

Which product investments actually move loyalty?

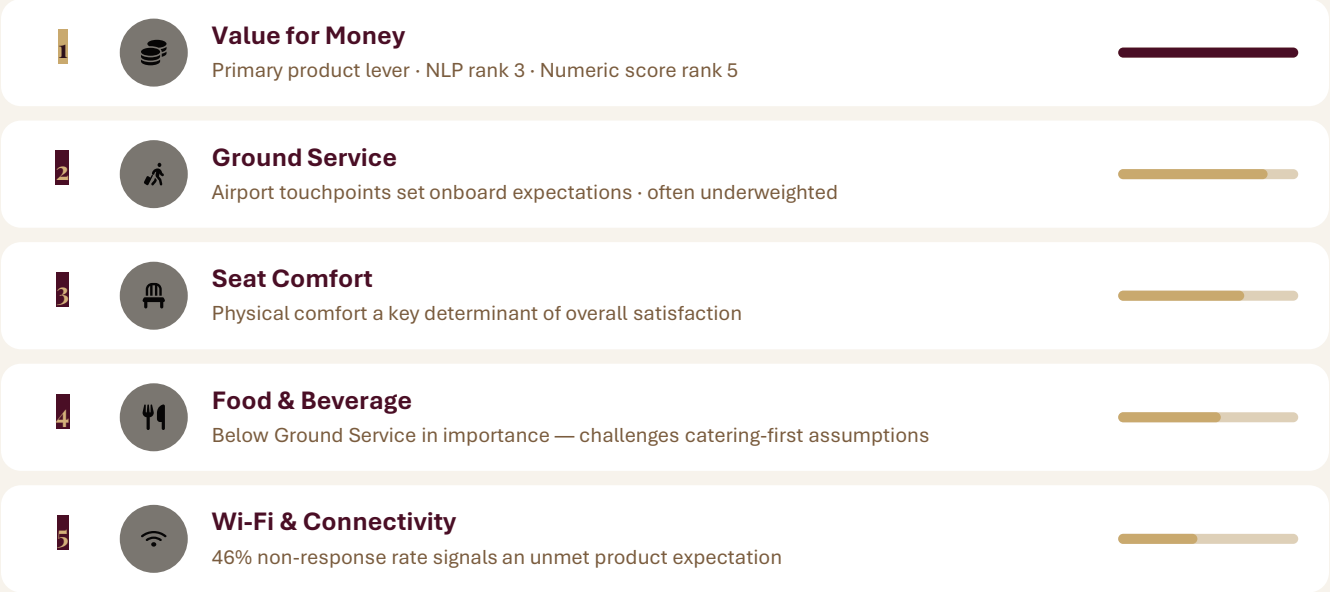
Model Performance



96%
Test accuracy across all models

SHAP decomposition per-passenger

Top 5 Loyalty Drivers — SHAP Feature Importance



Key insight: Ground Service ranks above Food & Beverage — airport touchpoints drive loyalty before boarding begins.

Section

02

Strategic Recommendations

Data-backed actions for product and commercial teams.

From analysis to action — five evidence-led product decisions, ranked by evidence strength.

Strategic Recommendations

Five data-backed product decisions — ranked by evidence strength.



Value for Money

Primary investment lever — product improvement must be made legible to passengers to move NPS.

NLP #1 Pearson r=0.86



Ground Service

Underweighted — airport touchpoints set expectations before boarding; low scores depress all onboard ratings.

Propensity #2 EDA



Economy Meal Service

A/B trial delivers high-confidence, statistically significant lift — roll out the treatment menu.

p<0.001 +0.30pt lift



IFE Refresh Cadence

Shift from bi-annual to quarterly — closes the recurring mid-year satisfaction dip at low marginal cost.

EDA Low cost



Captive Advocate Program

Identify passengers who recommend without loyalty before a competitor enters the route.

NLP SHAP

Technical Stack

End-to-end capability: from raw data to executive dashboard.

 Data & Analysis	Python 3.12 Pandas NumPy Matplotlib Seaborn SciPy
 Machine Learning	Scikit-learn XGBoost SHAP Logistic Regression Random Forest Stratified CV
 NLP & AI	Llama 3.1 8B Groq API Ollama Sentence Transformers all-MiniLM-L6-v2 Claude Code
 Clustering & Visualisation	K-Means UMAP Silhouette Analysis WordCloud
 BI & Deployment	Microsoft Power BI DAX Vercel Git GitHub BeautifulSoup Requests



Thank *You*



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